

Nepal Engineering College

Digital Signal Analysis & processing Tutorial 2

Starting Date:

Due Date:

- 1) Find Z-transform using the properties : (Mention the ROC)
 - a) $na^n u[n]$ b) $2^n u[n-2]$ c) $n^2 a^n u[n]$ d) $3(-\frac{1}{2})^n u[n] - 2(3)^n u[-n-1]$
 - e) $[3(2)^n - 4(3)^n] u[n]$ f) $a^n \sin(\omega_0 n) u[n]$
- 2) Find the system function $H(Z)$ and unit sample response $h[n]$ of the following system:
$$y[n] = \frac{1}{2} y[n-1] + 2x[n]$$
- 3) Determine the response of the system $y[n] = \frac{5}{6} y[n-1] - \frac{1}{6} y[n-2] + x[n]$ to the input signal,
 $x[n] = \delta[n] - \frac{1}{3} \delta[n-1]$.
- 4) Find inverse Z-transform of the following (use both partial fraction and long division method):
 - a) $X(z) = \frac{z^3}{(z+1)(z-1)^2}$, *find causal signal*
 - b) $X(z) = \frac{z}{3z^2 - 4z + 1}$, *find anticausal signal*
 - c) $X(z) = \frac{2 + 3z^{-1}}{(1+z^{-1})(1+\frac{1}{2}z^{-1})(1-\frac{1}{4}z^{-1})}$, *find causal signal*
- 5) Determine the step response of the causal system $y[n] = y[n-1] + x[n]$, is this stable system? Comment your answer.
- 6) A LTI system is described by the difference equation $y[n] = x[n] + 2x[n-1] + x[n-2]$
 - a Determine the impulse response of the system $h[n]$
 - b Is this stable system? Justify your answer.
 - c Determine frequency response of the system ie $H(e^{j\omega})$
 - d Sketch Magnitude and phase response of this system
- 7) When the input to an LTI system is $x[n] = (\frac{1}{2})^n u[n] + 2^n u[-n-1]$; the output is found to be
 $y[n] = 6(\frac{1}{2})^n u[n] - 6(\frac{3}{4})^n u[n]$.
 - a Find transfer function $H(z)$ of this system hence plot pole zero plot indicating ROC
 - b Find impulse response $h[n]$
 - c Write LCCD equation that characterizes the system
 - d Is the system stable? Is it Causal?
- 8) Consider the LTI system with transfer function $H(z) = \frac{z^{-2}}{(1 - \frac{1}{2}z^{-1})(1 - 3z^{-1})}$
 - a Suppose that the system is known to be stable, determine the output when the input is unit step sequence?
 - b Suppose the ROC of $H[z]$ includes $z=\infty$, Determine $y[n]$ evaluated at $n = 2$, when the input sequence is $x[n] = \{1, 2, 3\}$.
 - c Suppose we wish to recover $x[n]$ from $y[n]$ by processing $y[n]$ with a LTI system whose impulse response is given by $h_i[n]$ ie inverse of $h[n]$. Determine the response $h_i[n]$. Does $h_i[n]$ depends on ROC of $H(z)$?

(Note: You are advised to see all the examples from "Digital signal processing " by Proakis and Moanalkis and other available books.)